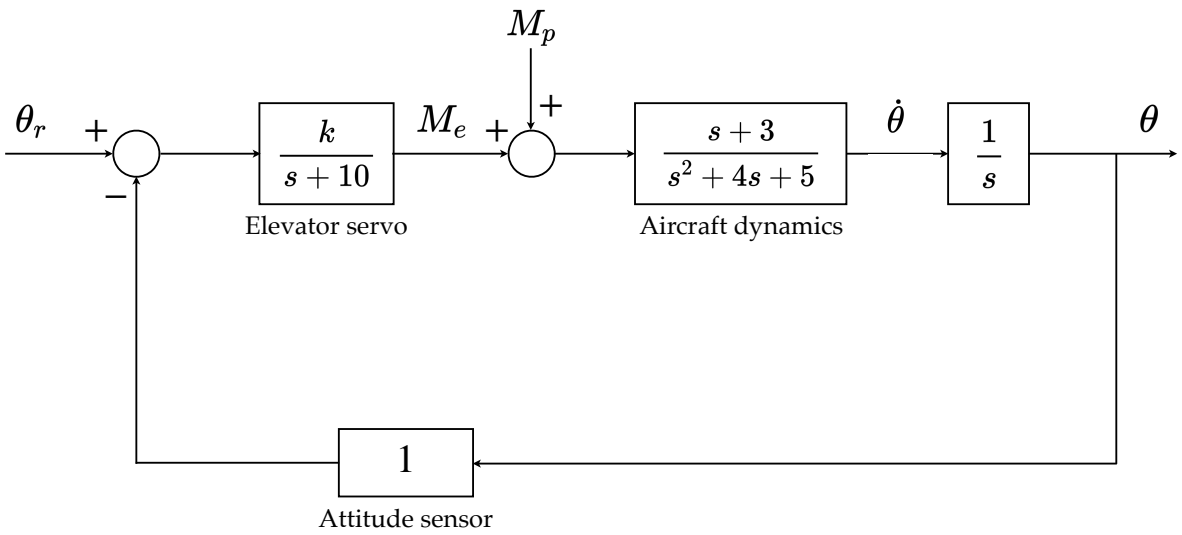
F26 AerE 3310: Flight Control Systems I

Practice Problem Set 7: Root Locus

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Problem 1 (Simplified version of exercise 5.39 from text)



Golden Nugget Airlines has opened a free bar in the tail of their airplanes in an attempt to lure customers. In order to automatically adjust for the sudden weight shift due to passengers rushing to the bar when it first opens, the airline is mechanizing a pitch-attitude autopilot shown above. We model the passenger moment as a step disturbance $M_p(s) = M_0/s$ with a maximum value of $M_0 = 0.1$ for a fully booked flight.

(a) Assuming the bar has opened and the passengers have rushed to it, what restriction on the value of k is required to keep the steady-state error in θ to less than 0.02 radians ($\approx 1^\circ$)? Show all your calculations.

Hint: Recall that you cannot apply FVT without ensuring the (necessary and sufficient conditions for) stability. The necessary and sufficient conditions for a degree four monic polynomial $s^4 + as^3 + bs^2 + cs + d$ to be stable is $a > 0, b > 0, c > 0, d > 0, abc - c^2 - a^2d > 0$.

Solution for Problem 1(a)

Following Lec. 17 p. 5 or PS06 Prob. 1(d), the disturbance-to-output transfer function is

$$\begin{aligned}\frac{\Theta(s)}{M_p(s)} &= \frac{\dfrac{s+3}{s^2+4s+5}\cdot\dfrac{1}{s}}{1+\dfrac{s+3}{s^2+4s+5}\cdot\dfrac{1}{s}\cdot\dfrac{k}{s+10}} \\ &= \frac{(s+3)(s+10)}{s(s+10)(s^2+4s+5)+k(s+3)} \\ &= \frac{(s+3)(s+10)}{s^4+14s^3+45s^2+(50+k)s+3k}.\end{aligned}$$

Applying the necessary and sufficient conditions of stability for the degree four polynomial, we get:

$$k>-50, \quad k>0, \quad 0>(50+k)^2-630(50+k)+(14)^2\cdot 3k.$$

The first two inequalities above imply $k>0$, while the last implies

$$0>k^2+58k-29000 \quad \Leftrightarrow \quad \underbrace{-7\sqrt{609}-29}_{\text{negative number}}<k<7\sqrt{609}-29.$$

So for stability (and thus to apply FVT), we need

$$0<k<7\sqrt{609}-29\approx 143.7455.$$

With k in the above range, the steady state value

$$\begin{aligned}\lim_{t\to\infty}\theta(t) &\overset{\text{(FVT)}}{=} \lim_{s\to 0}\, s\cdot\frac{\Theta(s)}{M_p(s)}\cdot M_p(s) \\ &= \lim_{s\to 0}\, s\cdot\frac{(s+3)(s+10)}{s^4+14s^3+45s^2+(50+k)s+3k}\cdot\frac{M_0}{s} \\ &= \frac{30M_0}{3k}=\frac{10M_0}{k}.\end{aligned}$$

Therefore, for the steady state error in θ to be less than 0.02 radians with the maximum (worst-case) $M_0=0.1$ (fully booked flight), we need

$$\frac{10\times 0.1}{k}<0.02=\frac{1}{50} \quad \Rightarrow \quad k>50,$$

which is more restrictive than what was found for stability considerations alone. Therefore, the k must satisfy:

$$50<k<7\sqrt{609}-29\approx 143.7455.$$

(b) Draw the root locus with respect to the gain k using MATLAB `rlocus` to get a visual estimate of the location on the pure imaginary axis where the system transitions from stable to unstable. You don't need to submit the MATLAB code. With the MATLAB visual intuition

in place, use your answer in part(a) to analytically derive this location on pure imaginary axis using hand calculations. Your hand calculations must be analytical. The MATLAB plot is only meant to help verify your hand calculations. It is okay to keep answer in terms of square root.

Solution for Problem 1(b)

Suppose the root locus crosses the imaginary axis at $\pm j\omega_0$ for some real number ω_0. Substituting $s = j\omega_0$ in the root locus polynomial, we obtain

$$(j\omega_0)^4 + 14(j\omega_0)^3 + 45(j\omega_0)^2 + (50+k)(j\omega_0) + 3k = 0.$$

Separating the real and imaginary parts of the above, we get

$$\omega_0^4 - 45\omega_0^2 + 3k = 0, \tag{1}$$

$$-14\omega_0^3 + (50+k)\omega_0 = 0. \tag{2}$$

We need to find common real root ω_0 for the above two equations. The roots of (2) are $\omega_0 = 0, \pm\sqrt{\dfrac{50+k}{14}}$. Out of these three, $\omega_0 = 0$ does not satisfy (1) since $k > 0$. Substituting $\omega_0 = \pm\sqrt{\dfrac{50+k}{14}}$ in (1) gives

$$\left(\frac{50+k}{14}\right)^2 - 45 \cdot \frac{50+k}{14} + 3k = 0 \quad \Rightarrow \quad (50+k)^2 - 630\,(50+k) + (14)^2 \cdot 3k = 0,$$

whose only positive root is

$$k_{\text{critical}} = 7\sqrt{609} - 29 \approx 143.7455,$$

precisely what appeared in our upper bound for k when applying the Routh-Hurwitz criterion in part (a). Hence the root locus crosses the pure imaginary axis at

$$\pm j\sqrt{\frac{50+k_{\text{critical}}}{14}} = \pm j\sqrt{\frac{21+7\sqrt{609}}{14}} \approx \pm j \cdot 3.7201.$$

These values match with the crossing locations observed via MATLAB `rlocus` for the loop transfer function $\dfrac{s+3}{s(s+10)(s^2+4s+5)}$.

Problem 2

Consider the loop transfer function

$$L(s) = \frac{s+1}{s(s+2)(s^2+2s+2)}.$$

Recall that the root locus is defined as the set

$$\{s \in \mathbb{C} \mid 1 + kL(s) = 0, 0 < k < \infty\},$$

i.e., the collection of complex numbers s that solve $1 + kL(s) = 0$ as we vary k from 0 to $+\infty$.

(a) Use the Routh-Hurwitz criterion to derive the value of $k \in (0, \infty)$ for which the root locus for $L(s)$ crosses the pure imaginary axis from left to right (i.e., the closed loop system transitions from stable to unstable). Use the same necessary and sufficient conditions given in the hint for Prob. 1(a). You can keep your answer in square roots and whole numbers, no need to compute decimals.

Solution for Problem 2(a)

The root locus polynomial is

$$1+kL(s)=0 \quad \Leftrightarrow \quad s(s+2)(s^2+2s+2)+k(s+1)=0 \quad \Leftrightarrow \quad s^4+4s^3+6s^2+(k+4)s+k=0.$$

Necessary conditions for stability gives

$$k > -4 \quad \text{and} \quad k > 0 \quad \Rightarrow \quad k > 0.$$

To determine the necessary and sufficient conditions, we apply the same necessary and sufficient conditions given in the hint for Prob. 1(a), to get the extra inequality:

$$24(k+4) - (k+4)^2 - 16k > 0 \quad \Rightarrow \quad 80 - k^2 > 0 \quad \Rightarrow \quad \left(4\sqrt{5}+k\right)\left(4\sqrt{5}-k\right) > 0,$$

which implies

$$\text{either} \quad 4\sqrt{5} > k > -4\sqrt{5}, \quad \text{or} \quad -4\sqrt{5} > k > 4\sqrt{5},$$

of which the second option (i.e., the inequality after "or") is impossible.
Because $20 > 4\sqrt{5}$ (equivalently, because $400 > 80$), we conclude that the necessary and sufficient conditions for stability is

$$4\sqrt{5} > k > 0.$$

Therefore, in the permissible range $(0, \infty)$ for k, the closed loop system transitions from being stable to unstable at $k = 4\sqrt{5}$.

(b) Use your answer in part (a) to derive the locations on the pure imaginary axis where this left-to-right crossing happens, i.e., determine the real number ω_0 such that the root locus crosses the pure imaginary axis at $\pm j\omega_0$ from left-to-right. You can keep your answer in square roots

and whole numbers, no need to compute decimals. Show all your hand calculations. You may want to verify your hand calculations with MATLAB `rlocus` command.

Solution for Problem 2(b)

To determine the left-to-right crossing location on the pure imaginary axis, we substitute $k = 4\sqrt{5}$ (using our answer to part(a)) and $s = j\omega_0$ in the root locus polynomial, to obtain

$$(j\omega_0)^4 + 4(j\omega_0)^3 + 6(j\omega_0)^2 + \left(4\sqrt{5}+4\right) j\omega_0 + 4\sqrt{5} = 0$$
$$\Rightarrow \quad \omega_0^4 + 4(-j)\omega_0^3 - 6\omega_0^2 + \left(4\sqrt{5}+4\right) j\omega_0 + 4\sqrt{5} = 0.$$

Equating the real and imaginary parts from both sides of the above, we get two equations

$$\omega_0^4 - 6\omega_0^2 + 4\sqrt{5} = 0, \tag{3}$$
$$-4\omega_0^3 + \left(4\sqrt{5}+4\right)\omega_0 = 0. \tag{4}$$

We need to find common real root ω_0 satisfying both (3) and (4). Equation (3) is a quadratic equation in ω_0^2, and yields

$$\begin{aligned}
\omega_0 &= \pm\sqrt{3 \pm \sqrt{9 - 4\sqrt{5}}} \\
&= \pm\sqrt{3 \pm \sqrt{\left(\sqrt{5}\right)^2 + (2)^2 - 2\cdot 2\cdot\sqrt{5}}} && \text{(completion of square)} \\
&= \pm\sqrt{3 \pm \sqrt{5} - 2} \\
&= \pm\sqrt{1 \pm \sqrt{5}}.
\end{aligned} \tag{5}$$

For the inner plus-minus in (5), we discard the inner minus since that would make ω_0 complex. Therefore, equation (3) gives real roots

$$\omega_0 = \pm\sqrt{1+\sqrt{5}}.$$

On the other hand, equation (4) gives three (all real) roots:

$$\omega_0\left(4\sqrt{5} + 4 - 4\omega_0^2\right) = 0 \quad \Rightarrow \quad \omega_0 = 0, \pm\sqrt{1+\sqrt{5}}.$$

Therefore, the common real roots for (3) and (4) are $\omega_0 = \pm\sqrt{1+\sqrt{5}}$.
Based on the above calculations, we conclude that the root locus for the given $L(s)$ crosses the pure imaginary axis from left-to-right at $\pm j\sqrt{1+\sqrt{5}} \approx \pm j1.7989$.